

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION
SSC PART-I (CLASS 9) PRACTICE EXAMINATION
CHEMISTRY PAPER 24

Syllabus: Chapter 11 – Environmental Chemistry: Air (35 marks) + Revision: Chapters 8, 10 (15 marks)

Note: Chapter 11 abhi class mein nahin parhaya gaya, ye paper aagay ka hai. Paper se pehle Quick Prep sheet zaroor parhein. Marks kam aayein to koi baat nahin, maqsad ye dekhna hai ke kya dobara parhna hai.

Total Marks: 50

Time Allowed: 2 hours 15 minutes

Version A

HIDAYAAT (Instructions)

1. **Saaray sawal** hal karnay hain. Is paper mein koi bhi **optional sawal nahin** hai.
2. Jawab apni notebook mein likhein. Har sawal ka number **bilkul waisa hi** likhein jaisa paper par hai, aur tarteeb se hal karein.
3. Section A mein sawal ka number likh kar sirf apnay option ka **letter** likhein, jaise **1. (b)**. Check kar lein ke jo letter aap ne likha hai, wohi option aap ne choose kiya hai.
4. Har equation **balance** karein, aur jahan **state symbols** maangay gaye hain wahan **(s), (l), (g), (aq)** zaroor likhein.
5. Oxygen gas ko hamesha **O₂** likhein, sirf O nahin.
6. Jahan **naam** maanga gaya ho wahan angrezi naam likhein, sirf formula nahin. Naam ya unit na likhne par $\frac{1}{2}$ **mark** katega.
7. Har explanation ke aakhir mein **ek mukammal jumla** likhein jo sawal ka jawab de.

SECTION A: MULTIPLE CHOICE QUESTIONS (10 × 1 = 10 MARKS)

#	Question	A	B	C	D	Answer
1	Approximately what percentage of clean, dry air is oxygen ?	78%	21%	1%	50%	(A)(B)(C)(D)
2	Besides acid rain, sulphur dioxide in the air is also responsible for	haze	photosynthesis	global cooling	melting of polar ice	(A)(B)(C)(D)
3	Particulate matter is harmful to health mainly because it can	turn rainwater acidic	form ozone in sunlight	penetrate deep into the lungs, causing respiratory problems and lung cancer	combine with haemoglobin in the blood	(A)(B)(C)(D)
4	According to your book, nearly half of all air pollution comes from	volcanoes	forest fires	termites and cows	cars and other motor vehicles	(A)(B)(C)(D)
5	The catalytic converter in a car contains metals such as	iron and nickel	platinum, palladium and rhodium	calcium oxide	copper and zinc	(A)(B)(C)(D)
6	Which one of these is an important source of methane ?	rice fields	car engines	lightning	power stations burning coal	(A)(B)(C)(D)
7	Photosynthesis removes which gas from the air?	carbon monoxide	sulphur dioxide	carbon dioxide	nitrogen dioxide	(A)(B)(C)(D)
8	Which equation correctly shows the ionization of nitric acid in water?	$H^+ + NO_3^- \rightarrow HNO_3$	$HNO_3 \rightarrow H^+ + NO_3^-$	$HNO_3 \rightarrow H_2 + NO_3$	$HNO_3 \rightleftharpoons H + NO_3$	(A)(B)(C)(D)
9	In the reaction $H_2S + NH_3 \rightleftharpoons NH_4^+ + HS^-$, the Bronsted-Lowry acid is	NH ₃	NH ₄ ⁺	HS ⁻	H ₂ S	(A)(B)(C)(D)
10	Oxygen gas exists as O ₂ molecules. The correct balanced equation for magnesium burning in oxygen is	$2Mg + O_2 \rightarrow 2MgO$	$Mg + O \rightarrow MgO$	$Mg + O_2 \rightarrow MgO$	$2Mg + O \rightarrow 2MgO$	(A)(B)(C)(D)

SECTION B: SHORT ANSWER QUESTIONS (6 × 4 = 24 MARKS)

Chhay ke chhay sawal hal karein. Har sawal 4 marks ka hai.

2.	(i) Define a pollutant. (1) (ii) Name two natural sources of air pollution and the pollutant that each one releases. (2) (iii) Name one human activity that causes air pollution, and the pollutants it releases. (1)	4
3.	Write a balanced equation, with state symbols , for the formation of each of these pollutants: (i) carbon monoxide, when carbon burns in a limited supply of oxygen (1) (ii) sulphur dioxide, from sulphur (1) (iii) nitrogen monoxide, from nitrogen and oxygen (1) (iv) nitrogen dioxide, from nitrogen monoxide (1)	4
4.	Carbon monoxide. (i) Under what condition is carbon monoxide produced when fuels burn? (1) (ii) Explain how carbon monoxide can cause death when it is inhaled. (2) (iii) Why can its presence in a room not be felt? (1)	4
5.	(i) Explain how greenhouse gases such as carbon dioxide and methane cause global warming. (2) (ii) State two things your book says will happen if global warming continues. (2)	4
6.	Ground-level ozone. (i) Why is ground-level ozone called a secondary pollutant? (1) (ii) Write the two steps, as equations, by which ozone forms in sunlight: first from nitrogen dioxide, then from the atomic oxygen produced. (2) (iii) Besides sunlight, name the two kinds of substances that react together to produce ground-level ozone. (1)	4
7.	(i) Draw one energy-profile diagram for an exothermic reaction, showing the pathway with and without a catalyst. Label both axes, the reactants, the products, the activation energy without the catalyst, the activation energy with the catalyst, and ΔH . (3) (ii) In one sentence, state what a catalyst does to the activation energy and what it does to ΔH . (1)	4

SECTION C: LONG ANSWER QUESTIONS (2 × 8 = 16 MARKS)

Dono sawal hal karein. Har sawal 8 marks ka hai.

8.	Acid rain. (i) State the pH of normal rainwater, and the pH below which rain is called acid rain. (2) (ii) Write balanced equations, with state symbols , to show how (a) sulphur dioxide is converted into sulphuric acid in the atmosphere (two equations), and (b) nitrogen dioxide forms nitric acid (one equation). (3) (iii) Write balanced equations for the action of acid rain on (a) iron, by sulphuric acid, and (b) marble, by nitric acid. (2) (iv) State two effects of acid rain on living things. (1)	8
9.	Bronsted-Lowry acids and bases, and naming. (i) Identify the Bronsted-Lowry acid and the Bronsted-Lowry base in each reaction, giving a reason for each: (4) (a) $\text{CH}_3\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}_3\text{O}^+$ (b) $\text{H}_2\text{O} + \text{NH}_3 \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$ (ii) Write the ionization equation, with charges, for (a) hydrochloric acid and (b) sulphuric acid in water. Use the correct arrow. (2) (iii) Write the name of each compound: $\text{Ca}(\text{OH})_2$, $\text{Mg}(\text{OH})_2$, Na_2O , MgO . (2)	8

END OF QUESTION PAPER

Chapters covered: Chapter 11 Environmental Chemistry: Air (MCQs 1–7, Questions 2 to 6 and 8) | Revision: Chapter 10 Acids, Bases and Salts (MCQs 8–10 and Question 9), Chapter 8 Energetics (Question 7).

Student's Name: _____

Date: _____

Marks Obtained: _____ / 50